

order check for separating T and U

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1 the problem

In our further quantum mechanics lecture notes [1], it was claimed that

$$e^{-\frac{i}{\hbar}H(t_n)\Delta t} e^{-\frac{i}{\hbar}H(t_{n-1})\Delta t} \dots e^{-\frac{i}{\hbar}H(t_1)\Delta t}$$

cannot be approximated as

$$e^{-\frac{i}{\hbar} \int_0^t H(t') dt'}$$

While the approximation

$$\prod_{i=n}^1 e^{-\frac{i}{\hbar}(T+V(t_i))\Delta t} \approx \prod_{i=n}^1 e^{-\frac{i\Delta t}{\hbar}T} e^{-\frac{i\Delta t}{\hbar}V(t_i)}$$

is valid.

However, I was confused when checking the orders.

it was concluded in the notes that

$$e^{-\frac{i}{\hbar}(T+V(t_i))\Delta t} = e^{-\frac{i\Delta t}{\hbar}T} e^{-\frac{i\Delta t}{\hbar}V(t_i)} + O(\Delta t^2)$$

, which is true.

However, it is also true that

$$e^{-\frac{i}{\hbar}(H(t_{i+1})+H(t_i))\Delta t} = e^{-\frac{i\Delta t}{\hbar}H(t_{i+1})} e^{-\frac{i\Delta t}{\hbar}H(t_i)} + O(\Delta t^2)$$

Given the same order of error, why the dividing the kinetic and potential energy terms is a valid approximation but neglecting the time-ordering is not?

2 solution: different global errors

Let us check the orders of the approximations carefully, starting with the naive removal of time ordering.

By using the BCH expansion [2]:

$$\prod_{i=n}^1 e^{-\frac{i\Delta t}{\hbar}H(t_i)} \approx e^{\left(\sum_i -\frac{i\Delta t}{\hbar}H(t_i) + \sum_{i \neq j} \frac{1}{2} \left(-\frac{i\Delta t}{\hbar}\right)^2 [H(t_i), H(t_j)] \right)}.$$

Let $N = \frac{t}{\Delta t}$, the number of commutator correction terms with $O(\Delta t^2)$ is given by the number of terms in the $i \neq j$ summation:

$$\binom{N}{2} = \frac{N(N-1)}{2} \sim O(\Delta t^{-2}),$$

so this global error does not disappear with $\Delta t \rightarrow 0$:

$$\boxed{\Delta U \sim O(1).}$$

On the other hand, although the local error of separating T and V is the same, the global error is different.

local error:

$$e^{-\frac{i\Delta t}{\hbar}(T+V(t_i))} = e^{-\frac{i\Delta t}{\hbar}T} e^{-\frac{i\Delta t}{\hbar}V(t_i)} + O(\Delta t^2),$$

expansion:

$$e^{-\frac{i\Delta t}{\hbar}(T+V(t_n))} \dots e^{-\frac{i\Delta t}{\hbar}(T+V(t_1))} \approx \\ \left(e^{-\frac{i\Delta t}{\hbar}T} e^{-\frac{i\Delta t}{\hbar}V(t_n)} + O(\Delta t^2) \right) \dots \left(e^{-\frac{i\Delta t}{\hbar}T} e^{-\frac{i\Delta t}{\hbar}V(t_1)} + O(\Delta t^2) \right).$$

still define $N = \frac{t}{\Delta t}$, since one bracket only contribute one error term, the global error is:

$$\boxed{O(\Delta t^2) \times N = O(\Delta t).}$$

This order-counting argument explains heuristically that the approximation will become precise with Δt substituted by dt . A rigorous justification requires a more careful analysis of the Baker–Campbell–Hausdorff expansion and operator convergence.

3 conclusion

In conclusion, although both approximations have local errors of order $O(\Delta t^2)$, their global errors scale differently. In the naive removal of time ordering, the number of commutator correction terms grows as $O(N^2)$ leading to a global error of order $O(1)$. In contrast, the T-V splitting accumulates only $O(N)$ local error terms, giving a global error of order $O(\Delta t)$. This explains why the latter becomes exact in the continuum limit whereas the former does not.

References

- [1] S. A. Parameswaran. *Further Quantum Mechanics*. Lecture notes, Oxford University. 2025.
- [2] Wikipedia contributors. *Baker–Campbell–Hausdorff formula*. 2026. URL: https://en.wikipedia.org/wiki/Baker%E2%80%93Campbell%E2%80%93Hausdorff_formula (visited on 06/30/2026).